

lim sup & lim inf

Def. Let $\{s_n\}$ be a sequence in $\mathbb{R}$. Let E be the set of all x (in the extended reals) such that $s_{n_k} \rightarrow x$ for some subsequence $\{s_{n_k}\}$. Then:

$$\limsup_{n \rightarrow \infty} s_n = \sup E, \quad \liminf_{n \rightarrow \infty} s_n = \inf E.$$

Equivalently: $\limsup s_n = \inf_n \sup_{k \geq n} s_k$ and $\liminf s_n = \sup_n \inf_{k \geq n} s_k$.

Thm. $s^* = \limsup s_n$ satisfies: (a) $s^* \in E$ (some subseq. converges to it); (b) If $x > s^*$, then $s_n < x$ for all but finitely many n . Unique with both properties.

$\lim s_n = s$ exists $\Leftrightarrow \limsup s_n = \liminf s_n = s$.

$s_n \leq t_n$ eventually $\Rightarrow \limsup s_n \leq \limsup t_n$ (same for $\liminf$).

Special Limits

$$\begin{array}{ll} 1/n^p \rightarrow 0 & p > 0 \\ \sqrt[n]{p} \rightarrow 1 & p > 0 \\ \sqrt[n]{n} \rightarrow 1 & \\ n^\alpha/(1+p)^n \rightarrow 0 & p > 0, \text{ any } \alpha \\ x^n \rightarrow 0 & |x| < 1 \end{array}$$

Metric Spaces & Topology

Metric Spaces

Def. Metric $d: X \times X \rightarrow \mathbb{R}$: (i) $d(x, y) \geq 0$, $d(x, y) = 0 \Leftrightarrow x = y$; (ii) $d(x, y) = d(y, x)$; (iii) $d(x, z) \leq d(x, y) + d(y, z)$. **Open ball**: $B(x, \varepsilon) = \{y : d(x, y) < \varepsilon\}$.

Key Definitions

Def. Open: every point is an interior point ($\exists \varepsilon$ -ball $\subseteq E$).

Closed: complement is open. Equiv: contains all its limit points.

Compact: every sequence in K has a subsequence converging to a point in K .

Connected: cannot be written as $A \cup B$ with A, B nonempty and separated ($\bar{A} \cap B = A \cap \bar{B} = \emptyset$, where $\bar{\cdot}$ = closure).

Dense: E is dense in X if every point of X is a limit point of E , or a point of E . Equiv: $\bar{E} = X$.

Perfect: closed with no isolated points ($E = E'$).

Bounded: $E \subseteq B(x, M)$ for some x, M .

Examples

Open	$(0, 1), B(x, r), \mathbb{R}, \emptyset$
Closed	$[0, 1], \{1/n\} \cup \{0\}, \mathbb{R}, \emptyset$
Compact	$[a, b], \{1, 1/2, \dots, 0\},$ finite sets
Not cpt	$(0, 1), \mathbb{R}, \mathbb{Z}$
Connected	intervals $[a, b], (a, b), \mathbb{R}$
Not conn	$[0, 1] \cup [2, 3], \mathbb{Q}$
Dense	$\mathbb{Q}$ in $\mathbb{R}$; irrationals in $\mathbb{R}$
Perfect	$[a, b],$ Cantor set C

Key Facts

$\bar{E} = E \cup E'$ (E' = limit points). E closed $\Leftrightarrow E' \subseteq E$. $p \in E' \Rightarrow$ every nbhd has ∞ -many points of E .

Compact $\Rightarrow$ closed & bounded. Closed $\subseteq$ compact $\Rightarrow$ compact.

Thm (Heine–Borel, $\mathbb{R}^n$ only). Closed & bounded $\Leftrightarrow$ compact $\Leftrightarrow$ every ∞ subset has limit point in E .

FIP: $\{K_\alpha\}$ compact, every finite $\cap$ nonempty $\Rightarrow \bigcap K_\alpha \neq \emptyset$. Weierstrass: bounded ∞ subset of $\mathbb{R}^n$ has a limit point.

$E \subset \mathbb{R}$ connected $\Leftrightarrow$ no gaps ($x, y \in E, x < z < y \Rightarrow z \in E$). Connected subsets of $\mathbb{R}$: intervals.

Cantor Set

Construction: $C = \bigcap C_n$ (remove middle thirds from $[0, 1]$). $C = \{x \in [0, 1] : \text{ternary expansion uses only } 0, 2\}$.

Properties: Closed ($\cap$ of closed), bounded $\Rightarrow$ compact. Perfect ($C' = C$). Uncountable ($C \sim \{0, 1\}^{\mathbb{N}}$). Nowhere dense (contains no interval). Measure zero ($\text{length}(C_n) = (2/3)^n \rightarrow 0$).

Uses: Uncountable set with measure 0; perfect set; totally disconnected compact set.

Continuity

Def. f continuous: V open $\Rightarrow f^{-1}(V)$ open. Equiv (ε - δ): $d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$.

How to show cts: (1) Direct ε - δ : bound $|f(x) - f(x_0)|$ in terms of $|x - x_0|$, choose δ . (2) Sequential: $x_n \rightarrow x_0 \Rightarrow f(x_n) \rightarrow f(x_0)$. (3) Algebraic: sums, products, quotients, compositions of cts functions are cts.

How to show NOT cts: Find $x_n \rightarrow x_0$ with $f(x_n) \not\rightarrow f(x_0)$, or two sequences approaching x_0 giving different limits.

Differentiability

$f'(x) = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}$. Exists $\Rightarrow$ diff'able at x .

$C^\infty \subset \dots \subset C^1 \subset C^0$. Diff'able $\Rightarrow$ cts (not vice versa).

To prove diff'able at x : Compute $\frac{f(x+h) - f(x)}{h}$ and show the limit as $h \rightarrow 0$ exists. Often needed at special points (e.g. $x = 0$).

Ex: $f(x) = x^2 \sin(1/x)$, $f(0) = 0$: $\frac{f(h)}{h} = h \sin(1/h) \rightarrow 0$, so $f'(0) = 0$. (Diff'able but f' not cts at 0: diff'able $\neq C^1$.)

To show NOT diff'able: Left and right limits of difference quotient disagree. Ex: $|x|$ at 0: left = -1 , right = 1 .

Uniform Continuity

Uniformly cts: $\forall \varepsilon > 0, \exists \delta > 0$: $d(p, q) < \delta \Rightarrow d(f(p), f(q)) < \varepsilon$ for ALL p, q . (One δ works everywhere.)

Cts on compact $\Rightarrow$ uniformly cts. Cts image of compact is compact; of connected is connected.

To show NOT unif. cts: Find x_n, y_n with $|x_n - y_n| \rightarrow 0$ but $|f(x_n) - f(y_n)| \not\rightarrow 0$. Ex: $f(x) = x^2$ on $\mathbb{R}$: $x_n = n, y_n = n + 1/n$, $|f(x_n) - f(y_n)| = 2 + 1/n^2 \rightarrow 2$.

Thm (Extreme Value). f cts on compact $K \Rightarrow$ attains max and min.

E non-compact $\Rightarrow \exists$ cts f : unbounded, or misses sup, or not unif. cts.

IVT and MVT

Thm (Intermediate Value). f cts on $[a, b]$, $f(a) < c < f(b) \Rightarrow \exists p \in (a, b)$ with $f(p) = c$.

Thm (Mean Value). f cts on $[a, b]$, diff'able on (a, b) : $\exists c$ with $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Thm (Generalized MVT). f, g cts on $[a, b]$, diff'able on (a, b) : $\exists c$ with

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Thm (L'Hôpital). $f, g \rightarrow 0$ as $x \rightarrow a$ and $\lim f'/g' = A \Rightarrow \lim f/g = A$.

Thm (IVT for Derivatives). $f'(a) < \lambda < f'(b) \Rightarrow \exists x \in (a, b)$ with $f'(x) = \lambda$.

Taylor's Theorem

Thm (Lagrange Remainder).

$$f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (\beta - \alpha)^k + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n$$

for some x between α and β .

Series

Convergence Tests

Thm (Comparison). $|a_n| \leq c_n$ and $\sum c_n$ converges $\Rightarrow \sum a_n$ converges.

Thm (Cauchy Condensation). $a_n \geq a_{n+1} \geq 0$: $\sum a_n$ conv. $\Leftrightarrow \sum 2^k a_{2^k}$ conv.

Thm (Root Test). $\alpha = \limsup |a_n|^{1/n}$. $\alpha < 1$: converges. $\alpha > 1$: diverges. $\alpha = 1$: inconclusive.

Thm (Ratio Test). $\limsup |a_{n+1}/a_n| < 1$: converges. $\liminf |a_{n+1}/a_n| > 1$: diverges. Root test is strictly stronger.

Absolute & Conditional Conv.

Def. $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. $\sum a_n$ converges conditionally if it converges but $\sum |a_n|$ diverges.

Thm. Absolute convergence $\Rightarrow$ convergence.

Example: $\sum (-1)^n/n$ converges conditionally.

Key: Absolutely convergent series can be rearranged freely. Conditionally convergent series can be rearranged to converge to any value (Riemann rearrangement theorem).

Thm (Alternating Series / Leibniz). $|c_1| \geq |c_2| \geq \dots$, alternating signs, $c_n \rightarrow 0 \Rightarrow \sum c_n$ converges.

Thm (Summation by Parts (Abel)).

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$$

where $A_n = \sum_{k=0}^n a_k$. Apply when $\{b_n\}$ is monotone.

Radius of Convergence

$\sum c_n z^n$: $R = 1/\limsup |c_n|^{1/n}$. Also: $R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ (when limit exists). Abs. conv. for $|z| < R$, diverges for $|z| > R$. If $|z| = R$: no general conclusion about absolute convergence (e.g. $\sum z^n/n$ conv. at $z = -1$ but not $z = 1$; $\sum z^n/n^2$ conv. at both). Unif.

conv. on $[-R + \varepsilon, R - \varepsilon]$. $f(x) = \sum c_n x^n$ is cts and $f'(x) = \sum n c_n x^{n-1}$ (same R).

$\sum 1/n^s$ conv. iff $s > 1$
 $\sum r^n$ conv. iff $|r| < 1$
 $\int_1^\infty x^{-p}$ conv. iff $p > 1$
 $\int_0^1 x^{-p}$ conv. iff $p < 1$

Which test to use: Factorials/ $n^n \rightarrow$ ratio. n th powers $\rightarrow$ root. Alternating $\rightarrow$ Leibniz. Looks like known series $\rightarrow$ comparison. Product $a_n b_n$ with oscillating/decaying $\rightarrow$ Abel.

Sequences/Series of Functions

Def. Pointwise: $f_n(x) \rightarrow f(x) \forall x$. **Uniform:** $\|f_n - f\|_\infty = \sup_x |f_n(x) - f(x)| \rightarrow 0$, i.e. single N works for ALL x .

Key difference: Pointwise: each x gets its own N . Uniform: one N works for every x at once (entire graph within ε -band).

Uniform $\Rightarrow$ pointwise. Converse fails: $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to $\mathbf{1}_{\{1\}}$ but not uniformly (limit discontinuous). To check uniform conv.: Compute $M_n = \sup_x |f_n(x) - f(x)|$ (find max via derivative). $M_n \rightarrow 0 \Rightarrow$ uniform.

To show NOT uniform: Find x_n with $|f_n(x_n) - f(x_n)| \not\rightarrow 0$, or show limit is discontinuous but each f_n is cts.

Thm (Cauchy Criterion). $\{f_n\}$ converges uniformly on $E \Leftrightarrow \forall \varepsilon > 0, \exists N: m, n \geq N \Rightarrow \|f_m - f_n\|_\infty < \varepsilon$.

$\mathcal{C}(X)$ with $\|\cdot\|_\infty$ is a complete metric space (X compact).

Thm (Weierstrass M-Test). $|f_n(x)| \leq M_n \forall x$ and $\sum M_n$ converges $\Rightarrow \sum f_n$ converges uniformly.

Uniform Conv. Interchange

+ Continuity (**Thm 7.12**)

Thm. $f_n \rightarrow f$ uniformly on E , each f_n cts $\Rightarrow f$ is cts.

Application: To show $\sum f_n$ converges to a cts function: (1) check each f_n is cts; (2) show $\sum f_n$ converges uniformly (e.g. M-test: $|f_n(x)| \leq M_n, \sum M_n < \infty$). Contrapositive: if the limit is discontinuous, convergence is NOT uniform.

Interchange of limits: $f_n \rightarrow f$ uniformly, $\lim_{t \rightarrow x} f_n(t) = A_n \Rightarrow \lim_{t \rightarrow x} f(t) = \lim_n A_n$.

+ Integration

Thm. $f_n \in \mathcal{R}[a, b], f_n \rightarrow f$ uniformly $\Rightarrow f \in \mathcal{R}[a, b]$ and

$$\int_a^b f dx = \lim_{n \rightarrow \infty} \int_a^b f_n dx.$$

Application: To compute $\int \sum f_n$: show uniform conv. (M-test), then swap: $\int \sum f_n = \sum \int f_n$.

+ Differentiation

Thm. $\{f_n\}$ diff'able on $[a, b], \{f_n(x_0)\}$ converges for some x_0 , and $f'_n \rightarrow g$ uniformly. Then $f_n \rightarrow f$ uniformly for some f , and $f' = g$. That is:

$$\left(\lim_{n \rightarrow \infty} f_n \right)' = \lim_{n \rightarrow \infty} f'_n.$$

Application: To differentiate a series term-by-term: (1) show $\sum f'_n$ converges uniformly; (2) check $\sum f_n(x_0)$ converges at one point. Then $(\sum f_n)' = \sum f'_n$.

Key: For differentiation, you need uniform convergence of the **derivatives** (not just the functions), plus pointwise convergence at one point.

Equicontinuity & Arzelà–Ascoli

Def. $\{f_n\} \subset \mathcal{C}(X)$ is **equicontinuous** on X if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ for **all** n and all x, y .

Ptwise bounded: $\forall x, \exists M_x: |f_n(x)| \leq M_x \forall n$. **Unif. bounded:** single M works $\forall n, x$.

Thm (Arzelà–Ascoli). K compact, $f_n \in \mathcal{C}(K)$, ptwise bounded + equicontinuous $\Rightarrow$ (a) uniformly bounded; (b) has uniformly convergent subsequence.

Application (“does $\{f_n\}$ have a unif. conv. subseq?”):

(1) K compact, each f_n cts.
 (2) Ptwise bounded: $|f_n(x)| \leq M_x \forall n$.
 (3) Equicontinuous: easiest via shared **Lipschitz bound**. If $|f'_n(x)| \leq L \forall n, x$, then $|f_n(x) - f_n(y)| \leq L|x - y|$ (MVT), so $\delta = \varepsilon/L$ works for all n .

Cite A-A $\Rightarrow$ done.

Def. f is **Lipschitz** with constant L if $|f(x) - f(y)| \leq L|x - y| \forall x, y$. Lipschitz $\Rightarrow$ unif. cts ($\delta = \varepsilon/L$). $|f'| \leq L \Rightarrow$ Lipschitz (MVT).

Stone–Weierstrass

Thm (Stone–Weierstrass). X compact metric space, $\mathcal{A} \subset \mathcal{C}(X)$ an algebra of real-valued cts functions. If $\mathcal{A}$ (1) **separates points** ($\forall x \neq y, \exists f \in \mathcal{A}: f(x) \neq f(y)$) and (2) **vanishes at no point** ($\forall x, \exists f \in \mathcal{A}: f(x) \neq 0$), then $\mathcal{A}$ is **dense** in $\mathcal{C}(X)$ w.r.t. $\|\cdot\|_\infty$.

Corollary (Weierstrass Approx): Every $f \in \mathcal{C}[a, b]$ is the uniform limit of polynomials. (Polynomials form an algebra separating points on $[a, b]$.)

Riemann Integrability

Setup

Partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$. $M_i = \sup_{[x_{i-1}, x_i]} f, m_i = \inf_{[x_{i-1}, x_i]} f$.

$$U(P, f) = \sum M_i \Delta x_i, \quad L(P, f) = \sum m_i \Delta x_i.$$

$$f \in \mathcal{R}[a, b] \text{ if } \overline{\int} f = \inf_P U(P, f) = \sup_P L(P, f) = \underline{\int} f.$$

Criterion

Thm. $f \in \mathcal{R}[a, b] \Leftrightarrow \forall \varepsilon > 0, \exists P$ with $U(P, f) - L(P, f) < \varepsilon$.

Sufficient Conditions

$f \in \mathcal{R}$ if: (i) continuous; (ii) monotone; (iii) bounded, finitely many disc'ties; (iv) bounded, disc'ties form a **zero set** ($\forall \varepsilon$, coverable by intervals of total length $< \varepsilon$).

Counterexamples

Thomae: $f(p/q) = 1/q, f(\text{irr.}) = 0$. Disc. set = $\mathbb{Q}$ (zero set) $\Rightarrow f \in \mathcal{R}$.

Dirichlet ($\mathbf{1}_{\mathbb{Q}}$): $U = 1, L = 0$ always $\Rightarrow f \notin \mathcal{R}$.

$1/\sqrt{x}$ on $(0, 1]$: unbounded $\Rightarrow \notin \mathcal{R}$ (improper integral exists).

Properties

Linearity. $f \leq g \Rightarrow \int f \leq \int g. |\int f| \leq \int |f|$.

Fundamental Theorem of Calculus

Thm (FTC I). $f \in \mathcal{R}[a, b]$. Define $F(x) = \int_a^x f(t) dt$. Then F is cts on $[a, b]$. If f is cts at x_0 , then F is diff'able at x_0 and $F'(x_0) = f(x_0)$.

Thm (FTC II). $f \in \mathcal{R}[a, b]$ and $\exists F$ diff'able on $[a, b]$ with $F' = f$. Then:

$$\int_a^b f(x) dx = F(b) - F(a).$$

Key Techniques

ε - δ **recipe:** (1) Write what you need. (2) Work backwards to express in terms of $d(p, q)$. (3) Read off δ .

Supremum argument: $E = \{t : t \text{ “too small”}\}$, $y = \sup E$. Show y too small and y too big both lead to contradictions.

Compactness trick: Local version at each point $\rightarrow$ open cover $\rightarrow$ finite subcover $\rightarrow$ take min/max over finitely many parameters.

Partition refinement: Isolate “bad” intervals into small total length. Bad: $\leq 2\|f\|_\infty \cdot$ (bad length). Good: $\leq \varepsilon(b - a)$.

Triple ε : For uniform conv. + continuity: $|f - f_n| + |f_n(\text{close})| + |f_n - f|$.

Diagonal argument: Extract successive subsequences, take diagonal. Used in Arzelà–Ascoli and uncountability proofs.

Quick Reference

Key Inequalities

- Cauchy–Schwarz: $|\langle u, v \rangle| \leq \|u\| \|v\|$
- Hölder: $\int |fg| \leq \|f\|_p \|g\|_q, \frac{1}{p} + \frac{1}{q} = 1$
- MVT bound: $|f(b) - f(a)| \leq \sup |f'| \cdot |b - a|$
- AM-GM: $\frac{a+b}{2} \geq \sqrt{ab}$

Convergence Summary

Uniform conv. preserves: continuity, integrability. For differentiation: need uniform conv. of **derivatives**.

$\mathbb{R}^n$ sets: closed + bounded $\Leftrightarrow$ compact
 Func. spaces: ptwise bdd + equicts $\Rightarrow$ conv. sub

Cauchy Criterion

$\{s_n\}$ converges $\Leftrightarrow \forall \varepsilon > 0, \exists N: m, n \geq N \Rightarrow |s_m - s_n| < \varepsilon$.

For series: $\sum a_n$ converges $\Rightarrow a_n \rightarrow 0$ (converse fails: harmonic).

Monotone + bounded $\Rightarrow$ convergent.